

SONATEL ACADEMIE

Formation Intelligence Artificielle • Promo 1



RAPPORT D'ATELIER LINUX

Préparation d'un environnement de développement IA

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Objectif

Mettre en place sous Linux un environnement complet de projet IA : arborescence, manipulation de fichiers, utilisateurs, groupes et permissions, installation des outils, téléchargement d'un dataset, gestion des processus, sauvegarde et automatisation Bash.

```
$ mkdir -p ~/IA_Project && cd ~/IA_Project
```

Le projet est créé dans le système de fichiers Linux (~/) et non sur /mnt/c : chmod et chown y sont sans effet.

Partie 1 — Organisation du projet

1.1 Création de l'arborescence

```
mkdir -p ~/IA_Project/{datasets/brut,config,logs,scripts,models}
cd ~/IA_Project
touch datasets/brut/{train.csv,test.csv,clients.csv,readme.txt}
touch config/{settings.conf,model.conf}
touch logs/{application.log,training.log}
touch scripts/{train.py,preprocess.py}
touch models/model_info.txt
```

1.2 Remplissage des fichiers

```
cat > datasets/brut/train.csv << 'EOF'
id,age,revenu,ville,achat
1,25,350000,Dakar,Oui
...
EOF
```

Idem pour test.csv, clients.csv, readme.txt, settings.conf, model.conf, application.log, training.log, train.py, preprocess.py, model_info.txt.

1.3 Vérification

```
tree ~/IA_Project
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ tree ~/IA_Project
/home/harouna/IA_Project
├── README.md
├── config
│   ├── model.conf
│   └── settings.conf
├── datasets
│   └── brut
│       ├── clients.csv
│       ├── readme.txt
│       ├── test.csv
│       └── train.csv
├── logs
│   ├── application.log
│   └── training.log
├── model.conf}
├── models
│   └── model_info.txt
├── preprocess.py
└── scripts
    ├── preprocess.py
    └── train.py

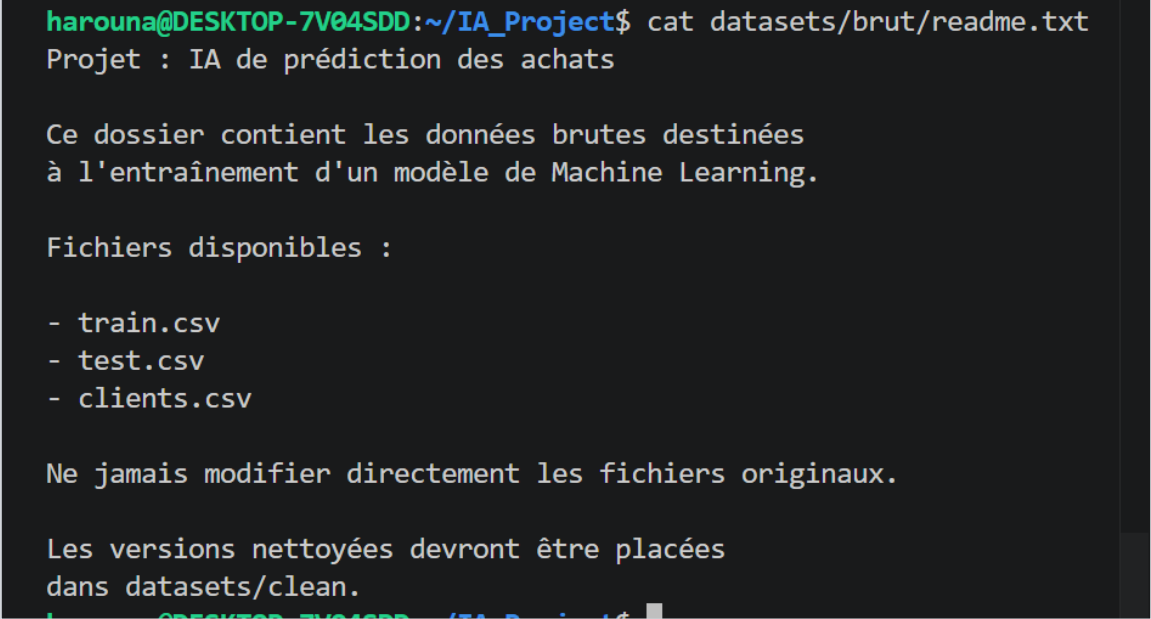
7 directories, 14 files
```

Capture 1 — Arborescence complète du projet (tree)

Partie 2 — Manipulation des fichiers

1) Afficher le contenu de readme.txt

```
cat datasets/brut/readme.txt
```



```
harouna@DESKTOP-7V04SDD:~/IA_Project$ cat datasets/brut/readme.txt
Projet : IA de prédiction des achats

Ce dossier contient les données brutes destinées
à l'entraînement d'un modèle de Machine Learning.

Fichiers disponibles :

- train.csv
- test.csv
- clients.csv

Ne jamais modifier directement les fichiers originaux.

Les versions nettoyées devront être placées
dans datasets/clean.
```

Capture 2 — Contenu de readme.txt

2) 5 premières lignes de train.csv | 3) 3 dernières lignes de application.log | 4) Nombre de lignes de clients.csv

```
head -n 5 datasets/brut/train.csv
tail -n 3 logs/application.log
wc -l datasets/brut/clients.csv
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ head -n 5 datasets/brut/train
.csv
id,age,revenu,ville,achat
1,25,350000,Dakar,Oui
2,42,720000,Thies,Non
3,31,500000,Saint-Louis,Oui
4,28,410000,Dakar,Oui
harouna@DESKTOP-7V04SDD:~/IA_Project$ tail -n 3 logs/application.log
2026-07-18 08:00:17 WARNING Valeurs manquantes détectées
2026-07-18 08:00:20 INFO Nettoyage terminé
2026-07-18 08:00:30 INFO Modèle prêt
harouna@DESKTOP-7V04SDD:~/IA_Project$ wc -l datasets/brut/clients.csv
6 datasets/brut/clients.csv
```

Capture 3 — head, tail et wc -l

5) Lignes contenant WARNING | 6) Le mot INFO dans les journaux

```
grep "WARNING" logs/application.log
grep -rn "WARNING" .
grep "INFO" logs/*.log
grep -c "INFO" logs/application.log
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ grep "WARNING" logs/application.log
2026-07-18 08:00:17 WARNING Valeurs manquantes détectées
harouna@DESKTOP-7V04SDD:~/IA_Project$ grep "INFO" logs/*.log
logs/application.log:2026-07-18 08:00:10 INFO Application démarrée
logs/application.log:2026-07-18 08:00:15 INFO Chargement du dataset
logs/application.log:2026-07-18 08:00:20 INFO Nettoyage terminé
logs/application.log:2026-07-18 08:00:30 INFO Modèle prêt
harouna@DESKTOP-7V04SDD:~/IA_Project$
```

Capture 4 — Recherches grep : WARNING et INFO

7) Copier train.csv dans datasets/clean | 8) Renommer | 9) Supprimer test.csv

```
mkdir -p datasets/clean
cp datasets/brut/train.csv datasets/clean/
mv models/model_info.txt models/modele.txt
rm datasets/brut/test.csv
ls -l datasets/clean models datasets/brut
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ mkdir -p datasets/clean
harouna@DESKTOP-7V04SDD:~/IA_Project$ cp datasets/brut/train.csv da
tasetes/clean/
harouna@DESKTOP-7V04SDD:~/IA_Project$ ls -l datasets/clean/
total 4
-rw-r--r-- 1 harouna harouna 260 Jul 28 12:37 train.csv
harouna@DESKTOP-7V04SDD:~/IA_Project$ mv models/model_info.txt mode
ls/modele.txt
harouna@DESKTOP-7V04SDD:~/IA_Project$ ls -l models/
total 4
-rw-r--r-- 1 harouna harouna 98 Jul 28 12:06 modele.txt
harouna@DESKTOP-7V04SDD:~/IA_Project$ rm datasets/brut/test.csv
harouna@DESKTOP-7V04SDD:~/IA_Project$ ls -l datasets/brut/
total 12
-rw-r--r-- 1 harouna harouna 121 Jul 28 11:53 clients.csv
-rw-r--r-- 1 harouna harouna 330 Jul 28 11:58 readme.txt
-rw-r--r-- 1 harouna harouna 260 Jul 28 11:21 train.csv
```

Capture 5 — cp, mv, rm et vérification par ls -l

10) Fichiers .csv | 11) Fichiers .log

```
find ~/IA_Project -type f -name "*.csv"
find ~/IA_Project -type f -name "*.log"
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ find ~/IA_Project -type f -na
me "*.csv"
/home/harouna/IA_Project/datasets/clean/train.csv
/home/harouna/IA_Project/datasets/brut/train.csv
/home/harouna/IA_Project/datasets/brut/clients.csv
harouna@DESKTOP-7V04SDD:~/IA_Project$ find ~/IA_Project -type f -na
me "*.log"
/home/harouna/IA_Project/logs/application.log
/home/harouna/IA_Project/logs/training.log
harouna@DESKTOP-7V04SDD:~/IA_Project$
```

Capture 6 — Recherche des fichiers .csv et .log (find)

12) Fichiers triés par taille

```
ls -lhS
du -ah ~/IA_Project | sort -rh | head -20
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ ls -lhS
total 24K
drwxr-xr-x 2 harouna harouna 4.0K Jul 28 12:04 config
drwxr-xr-x 4 harouna harouna 4.0K Jul 28 12:37 datasets
drwxr-xr-x 2 harouna harouna 4.0K Jul 28 12:05 logs
drwxr-xr-x 2 harouna harouna 4.0K Jul 28 12:38 models
drwxr-xr-x 2 harouna harouna 4.0K Jul 28 12:12 scripts
-rw-r--r-- 1 harouna harouna 2.9K Jul 27 15:52 README.md
-rw-r--r-- 1 harouna harouna  0 Jul 28 11:16 model.conf}
-rw-r--r-- 1 harouna harouna  0 Jul 28 11:19 preprocess.py
```

Capture 7 — Fichiers triés par taille

13) Concaténer les logs | 14) Extraire les lignes Loss

```
cat logs/application.log logs/training.log > logs/all_logs.log
grep "Loss" logs/training.log > logs/loss.log
cat logs/loss.log
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ cat logs/application.log logs
/training.log > logs/all_logs.log
harouna@DESKTOP-7V04SDD:~/IA_Project$ cat logs/all_logs.log
2026-07-18 08:00:10 INFO Application démarrée
2026-07-18 08:00:15 INFO Chargement du dataset
2026-07-18 08:00:17 WARNING Valeurs manquantes détectées
2026-07-18 08:00:20 INFO Nettoyage terminé
2026-07-18 08:00:30 INFO Modèle prêt
Epoch 1 : Loss = 0.82
Epoch 2 : Loss = 0.60
Epoch 3 : Loss = 0.45
Epoch 4 : Loss = 0.32
Epoch 5 : Loss = 0.28
Training terminé.
harouna@DESKTOP-7V04SDD:~/IA_Project$ grep "Loss" logs/training.log
> logs/loss.log
harouna@DESKTOP-7V04SDD:~/IA_Project$ cat logs/loss.log
Epoch 1 : Loss = 0.82
Epoch 2 : Loss = 0.60
Epoch 3 : Loss = 0.45
Epoch 4 : Loss = 0.32
Epoch 5 : Loss = 0.28
harouna@DESKTOP-7V04SDD:~/IA_Project$
```

Capture 8 — Concaténation des logs et création de loss.log

15) Nombre de fichiers du projet

```
find ~/IA_Project -type f | wc -l  
find ~/IA_Project -type d | wc -l
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ find -type f | wc -l  
73
```

Capture 9 — Comptage des fichiers et dossiers

16) Sauvegarde backup/projet_IA.tar.gz

```
mkdir -p ~/IA_Project/backup  
cd ~  
tar -czvf ~/IA_Project/backup/projet_IA.tar.gz --exclude='IA_Project/backup'  
IA_Project  
ls -lh ~/IA_Project/backup/
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ cd ~
harouna@DESKTOP-7V04SDD:~$ tar -czvf ~/IA_Project/backup/projet_IA.
tar.gz --exclude='IA_Project/backup' IA_Project
IA_Project/
IA_Project/datasets/
IA_Project/datasets/clean/
IA_Project/datasets/clean/train.csv
IA_Project/datasets/brut/
IA_Project/datasets/brut/train.csv
IA_Project/datasets/brut/readme.txt
IA_Project/datasets/brut/clients.csv
IA_Project/README.md
IA_Project/preprocess.py
IA_Project/models/
IA_Project/models/modele.txt
IA_Project/logs/
IA_Project/logs/application.log
IA_Project/logs/all_logs.log
IA_Project/logs/training.log
IA_Project/logs/loss.log
IA_Project/.git/
IA_Project/.git/COMMIT_EDITMSG
IA_Project/.git/index
IA_Project/.git/description
```

```
harouna@DESKTOP-7V04SDD:~$ ls -lh ~/IA_Project/backup/
total 24K
-rw-r--r-- 1 harouna harouna 22K Jul 28 13:19 projet_IA.tar.gz
harouna@DESKTOP-7V04SDD:~$
```

Capture 10 — Création de l'archive projet_IA.tar.gz

17) Restauration hors du projet

```
mkdir -p ~/restauration
tar -xzvf ~/IA_Project/backup/projet_IA.tar.gz -C ~/restauration
tree ~/restauration
```

```
harouna@DESKTOP-7V04SDD:~$ mkdir -p ~/restauration
harouna@DESKTOP-7V04SDD:~$ tar -xzf ~/IA_Project/backup/projet_IA.
tar.gz -C ~/restauration
IA_Project/
IA_Project/datasets/
IA_Project/datasets/clean/
IA_Project/datasets/clean/train.csv
IA_Project/datasets/brut/
IA_Project/datasets/brut/train.csv
IA_Project/datasets/brut/readme.txt
IA_Project/datasets/brut/clients.csv
IA_Project/README.md
IA_Project/preprocess.py
IA_Project/models/
IA_Project/models/modele.txt
IA_Project/logs/
IA_Project/logs/application.log
IA_Project/logs/all_logs.log
IA_Project/logs/training.log
IA_Project/logs/loss.log
IA_Project/.git/
IA_Project/.git/COMMIT_EDITMSG
IA_Project/.git/index
IA_Project/.git/description
IA_Project/.git/branches/
IA_Project/.git/refs/
IA_Project/.git/refs/tags/
IA_Project/.git/refs/heads/
```

```
harouna@DESKTOP-7V04SDD:~$ tree ~/restauration
/home/harouna/restauration
├── IA_Project
│   ├── README.md
│   ├── config
│   │   ├── model.conf
│   │   └── settings.conf
│   ├── datasets
│   │   ├── brut
│   │   │   ├── clients.csv
│   │   │   ├── readme.txt
│   │   │   └── train.csv
│   │   └── clean
│   │       └── train.csv
│   ├── logs
│   │   ├── all_logs.log
│   │   ├── application.log
│   │   ├── loss.log
│   │   └── training.log
│   ├── models
│   │   └── modele.txt
│   ├── preprocess.py
│   └── scripts
│       ├── preprocess.py
│       └── train.py
9 directories, 15 files
harouna@DESKTOP-7V04SDD:~$
```

Capture 11 — Restauration de la sauvegarde dans ~/restauration

Partie 3 — Utilisateurs et permissions

1) Création des utilisateurs

```
sudo adduser --gecos "Alice,,,Data Scientist" alice
sudo adduser --gecos "Bob,,,Data Engineer" bob
sudo adduser --gecos "Charles,,,Developpeur API" charles
sudo adduser --gecos "Diane,,,MLOps Engineer" diane
sudo adduser --gecos "Eva,,,Stagiaire" eva
```

```
harouna@DESKTOP-7V04SDD:~$ sudo adduser --gecos "Alice,,,Data Scientist" alice
[sudo] password for harouna:
info: Adding user `alice' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `alice' (1002) ...
info: Adding new user `alice' (1002) with group `alice (1002)' ...
info: Creating home directory `/home/alice' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
info: Adding new user `alice' to supplemental / extra groups `users'
...
info: Adding user `alice' to group `users' ...
```

```
harouna@DESKTOP-7V04SDD:~$ sudo adduser --gecos "Bob,,,Data Engineer" bob
info: Adding user `bob' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `bob' (1003) ...
info: Adding new user `bob' (1003) with group `bob (1003)' ...
info: Creating home directory `/home/bob' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
info: Adding new user `bob' to supplemental / extra groups `users'
...
info: Adding user `bob' to group `users' ...
harouna@DESKTOP-7V04SDD:~$ sudo adduser --gecos "Charles,,,Développeur API" charles
info: Adding user `charles' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `charles' (1004) ...
info: Adding new user `charles' (1004) with group `charles (1004)'
...
info: Creating home directory `/home/charles' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
info: Adding new user `charles' to supplemental / extra groups `users'
...
```

```
harouna@DESKTOP-7V04SDD:~$ sudo adduser --gecos "Diane,,,MLOps Engi
neer" diane
info: Adding user `diane' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `diane' (1005) ...
info: Adding new user `diane' (1005) with group `diane (1005)' ...
info: Creating home directory `/home/diane' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
info: Adding new user `diane' to supplemental / extra groups `users'
...
info: Adding user `diane' to group `users' ...
harouna@DESKTOP-7V04SDD:~$ sudo adduser --gecos "Eva,,,Stagiaire" e
va
info: Adding user `eva' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `eva' (1006) ...
info: Adding new user `eva' (1006) with group `eva (1006)' ...
info: Creating home directory `/home/eva' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
info: Adding new user `eva' to supplemental / extra groups `users'
...
info: Adding user `eva' to group `users' ...
```

Capture 12 — Création des comptes utilisateurs

Vérification des répertoires personnels, UID et groupes

```
ls -ld /home/alice /home/bob /home/charles /home/diane /home/eva
for u in alice bob charles diane eva; do echo "--- $u ---"; id $u; groups $u;
done
```

```
harouna@DESKTOP-7V04SDD:~$ ls -ld /home/alice /home/bob /home/charl
es /home/diane /home/eva
drwxr-x--- 2 alice  alice  4096 Jul 28 14:55 /home/alice
drwxr-x--- 2 bob    bob    4096 Jul 28 14:59 /home/bob
drwxr-x--- 2 charles charles 4096 Jul 28 15:01 /home/charles
drwxr-x--- 2 diane  diane  4096 Jul 28 15:04 /home/diane
drwxr-x--- 2 eva    eva    4096 Jul 28 15:05 /home/eva
harouna@DESKTOP-7V04SDD:~$
```

```
harouna@DESKTOP-7V04SDD:~$ for u in alice bob charles diane eva; do
    echo "--- $u ---"
    id "$u"
    groups "$u"
done
--- alice ---
uid=1002(alice) gid=1002(alice) groups=1002(alice),100(users)
alice : alice users
--- bob ---
uid=1003(bob) gid=1003(bob) groups=1003(bob),100(users)
bob : bob users
--- charles ---
uid=1004(charles) gid=1004(charles) groups=1004(charles),100(users)
charles : charles users
--- diane ---
uid=1005(diane) gid=1005(diane) groups=1005(diane),100(users)
diane : diane users
--- eva ---
uid=1006(eva) gid=1006(eva) groups=1006(eva),100(users)
eva : eva users
harouna@DESKTOP-7V04SDD:~$
```

Capture 13 — Répertoires personnels, UID et groupes

2) Création des groupes

```
sudo groupadd data
sudo groupadd models
sudo groupadd api
sudo groupadd mlops
sudo groupadd interns
grep -E "^(data|models|api|mlops|interns):" /etc/group
```

```
harouna@DESKTOP-7V04SDD:~$ sudo groupadd data
harouna@DESKTOP-7V04SDD:~$ sudo groupadd models
harouna@DESKTOP-7V04SDD:~$ sudo groupadd api
harouna@DESKTOP-7V04SDD:~$ sudo groupadd mlops
harouna@DESKTOP-7V04SDD:~$ sudo groupadd interns
harouna@DESKTOP-7V04SDD:~$ echo groups
groups
harouna@DESKTOP-7V04SDD:~$ groups
harouna adm cdrom sudo dip plugdev users docker
harouna@DESKTOP-7V04SDD:~$ grep -E "^(data|models|api|mlops|interns
):" /etc/group
data:x:1007:
models:x:1008:
api:x:1009:
mlops:x:1010:
interns:x:1011:
harouna@DESKTOP-7V04SDD:~$
```

Capture 14 — Groupes créés (/etc/group)

3) Affectation des utilisateurs aux groupes

```
sudo usermod -aG data,models alice
sudo usermod -aG data bob
sudo usermod -aG api charles
sudo usermod -aG mlops diane
sudo usermod -aG interns eva
for u in alice bob charles diane eva; do id $u; done
```

-aG : ajoute sans effacer les groupes existants.

```
harouna@DESKTOP-7V04SDD:~$ sudo usermod -aG data,models alice
harouna@DESKTOP-7V04SDD:~$ sudo usermod -aG data bob
harouna@DESKTOP-7V04SDD:~$ sudo usermode -aG api charles
sudo: usermode: command not found
harouna@DESKTOP-7V04SDD:~$ sudo usermod -aG api charles
harouna@DESKTOP-7V04SDD:~$ sudo usermod -aG mlops diane
harouna@DESKTOP-7V04SDD:~$ sudo usermod -aG interns eva
harouna@DESKTOP-7V04SDD:~$ for u in alice bob charles diane eva; do
  id "$u"; done
uid=1002(alice) gid=1002(alice) groups=1002(alice),100(users),1007(
data),1008(models)
uid=1003(bob) gid=1003(bob) groups=1003(bob),100(users),1007(data)
uid=1004(charles) gid=1004(charles) groups=1004(charles),100(users)
,1009(api)
uid=1005(diane) gid=1005(diane) groups=1005(diane),100(users),1010(
mlops)
uid=1006(eva) gid=1006(eva) groups=1006(eva),100(users),1011(intern
s)
```

Capture 15 — Utilisateurs rattachés à leurs groupes (id)

4) Sous-dossiers à la racine du projet

```
cd ~/IA_Project
mkdir -p api backup documentation shared
chmod o+x /home/$USER
chmod 755 ~/IA_Project
ls -l ~/IA_Project
```

```
harouna@DESKTOP-7V04SDD:~$ ls -l
total 167048
drwxr-xr-x  9 harouna harouna      4096 Jul 28 13:09 IA_Project
drwxr-xr-x  3 root    root      4096 Jul 16  2025 ProjetStorm
drwxr-xr-x  8 harouna harouna      4096 Jul 21  2025 ProjetStormLin
ux
drwxr-xr-x 13 harouna harouna      4096 Jul 17  2025 apache-storm-2
.8.1
-rw-r--r--  1 harouna harouna 171007675 Jul 17  2025 apache-storm-2
.8.1.tar.gz
drwxr-xr-x  2 harouna harouna      4096 Jul 28 15:35 api
drwxr-xr-x  2 harouna harouna      4096 Jul 28 15:35 backup
drwxr-xr-x  2 harouna harouna      4096 Jul 28 15:35 documentation
-rw-r--r--  1 harouna harouna       308 Jul 24 12:07 nettoyage_2026
0724_120749.log
drwxr-xr-x  2 harouna harouna      4096 Jul 24 12:07 projet_ia
drwxr-xr-x  3 harouna harouna      4096 Jul 28 13:22 restauration
drwxr-xr-x  2 harouna harouna      4096 Jul 28 15:35 shared
drwxr-xr-x  2 harouna harouna      4096 Jul 17  2025 tools
harouna@DESKTOP-7V04SDD:~$
```

Capture 16 — Sous-dossiers api, backup, documentation, shared

5) et 6) Propriétaires et groupes

```
sudo chown -R alice:data datasets
sudo chown -R alice:data models
sudo chown -R charles:api api
sudo chown -R diane:mlops logs
sudo chown -R diane:mlops backup
sudo chown -R bob:data documentation
sudo chown -R root:data shared
ls -l ~/IA_Project
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R charles api
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R alice datasets
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R alice models
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R charles api
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R diane logs
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R diane backup
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R bob documentati
on
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R root shared
chown: cannot access 'shared': No such file or directory
harouna@DESKTOP-7V04SDD:~/IA_Project$ cd ~
harouna@DESKTOP-7V04SDD:~$ mv shared ~/IA_Project
harouna@DESKTOP-7V04SDD:~$ cd ~/IA_Project
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chown -R root shared
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R data datasets
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R data models
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R api api
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R mllops logs
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R mllops backup
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R data documentat
ion
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R data shared
harouna@DESKTOP-7V04SDD:~/IA_Project$
```

Capture 17 — Propriétaires et groupes attribués (ls -l)

7) Configuration des droits

Dossier	Propriétaire	Groupe	Autres	Octal
datasets	rwX	rwX	---	770
models	rwX	r-X	---	750
api	rwX	rwX	---	770
logs	rwX	r--	---	740
backup	rwX	---	---	700
documentation	rwX	r--	r--	744
shared	rwX	rwX	r-X	775

Tableau 1 — Matrice des droits appliqués aux dossiers du projet

```
cd ~/IA_Project
sudo chmod -R 770 datasets
sudo chmod -R 750 models
sudo chmod -R 770 api
```

```

sudo chmod -R 740 logs
sudo chmod -R 700 backup
sudo chmod -R 744 documentation
sudo chmod -R 775 shared
ls -l ~/IA_Project

```

```

harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R data models
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R api api
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R mlops logs
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R mlops backup
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R data documentat
ion
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chgrp -R data shared
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chmod -R 770 datasets
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chmod -R 750 models
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chmod -R 770 api
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chmod -R 740 logs
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chmod -R 700 backup
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chmod -R 744 documentati
on
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo chmod -R 775 shared
harouna@DESKTOP-7V04SDD:~/IA_Project$ ls -l ~/IA/_Project
ls: cannot access '~/IA/_Project': No such file or directory
harouna@DESKTOP-7V04SDD:~/IA_Project$ ls -l
total 40
-rw-r--r-- 1 harouna harouna 2903 Jul 27 15:52 README.md
drwxrwx--- 2 charles api 4096 Jul 28 15:35 api
drwx----- 2 diane mlops 4096 Jul 28 13:15 backup
drwxr-xr-x 2 harouna harouna 4096 Jul 28 12:04 config
drwxrwx--- 4 alice data 4096 Jul 28 12:37 datasets
drwxr--r-- 2 bob data 4096 Jul 28 15:35 documentation
drwxr----- 2 diane mlops 4096 Jul 28 12:51 logs
drwxr-x--- 2 alice data 4096 Jul 28 12:38 models
-rw-r--r-- 1 harouna harouna 0 Jul 28 11:19 preprocess.py
drwxr-xr-x 2 harouna harouna 4096 Jul 28 12:12 scripts

```

Capture 18 — Permissions appliquées (ls -l de la racine du projet)

8) Vérification des droits depuis chaque compte

```

# ALICE : peut créer un fichier dans datasets
sudo -u alice touch ~harouna/IA_Project/datasets/test_alice.txt

# BOB : lit datasets, ne modifie pas models
sudo -u bob ls ~harouna/IA_Project/datasets
sudo -u bob touch ~harouna/IA_Project/models/interdit.txt

```

```

harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u alice bash -c 'touch
~harouna/IA_Project/datasets/test_alice.txt && echo "OK alice dans
datasets"'
OK alice dans datasets
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u bob ls ~harouna/IA_Pr
oject/datasets
brut clean test_alice.txt
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u bob touch ~harouna/IA
_Project/models/interdit.txt
touch: cannot touch '/home/harouna/IA_Project/models/interdit.txt':
Permission denied
harouna@DESKTOP-7V04SDD:~/IA_Project$

```

Capture 19 — Tests des droits : alice et bob

```

# CHARLES : travaille dans api
sudo -u charles touch ~harouna/IA_Project/api/routes.py
# DIANE : propriétaire de logs et backup
sudo -u diane touch ~harouna/IA_Project/logs/deploy.log
# EVA : lit documentation, pas d'accès à backup
sudo -u eva ls ~harouna/IA_Project/documentation
sudo -u eva ls ~harouna/IA_Project/backup

```

```

touch: cannot touch '/home/harouna/IA_Project/models/interdit.txt': Permission denied
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u charles touch ~harouna/IA_Project/api/routes.
py
[sudo] password for harouna:
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u charles touch ~harouna/IA_Project/api/routes.
py
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u charles ls ~harouna/I
A_Project/models
ls: cannot open directory '/home/harouna/IA_Project/models': Permission denied
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u diane touch ~harouna/
IA_Project/logs/deploy.log
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u diane ls ~harouna/IA_Project/backup
projet_IA.tar.gz
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u eva ls ~harouna/IA_Project/documentation
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u eva cd ~harouna/IA_Pr
oject/documentation
sudo: cd: command not found
sudo: "cd" is a shell built-in command, it cannot be run directly.
sudo: the -s option may be used to run a privileged shell.
sudo: the -D option may be used to run a command in a specific directory.
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u eva ls ~harouna/IA_Pr
oject/backup
ls: cannot open directory '/home/harouna/IA_Project/backup': Permission denied
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u eva touch ~harouna/IA
_Project/backup/x
touch: cannot touch '/home/harouna/IA_Project/backup/x': Permission denied
harouna@DESKTOP-7V04SDD:~/IA_Project$

```

Capture 20 — Tests des droits : charles, diane et eva

9) Intégration de Frank (Data Scientist)

```
# a) compte
sudo adduser --gecos "Frank,,Data Scientist" frank
# b) groupes
sudo usermod -aG data,models frank
id frank
# c) peut travailler dans datasets/
sudo -u frank touch ~harouna/IA_Project/datasets/frank_notes.txt
# d) ne peut pas modifier backup/
sudo -u frank touch ~harouna/IA_Project/backup/test.txt
# e) peut accéder à shared/
sudo -u frank bash -c 'cd ~harouna/IA_Project/shared && touch frank_shared.txt
&& ls -l'
```

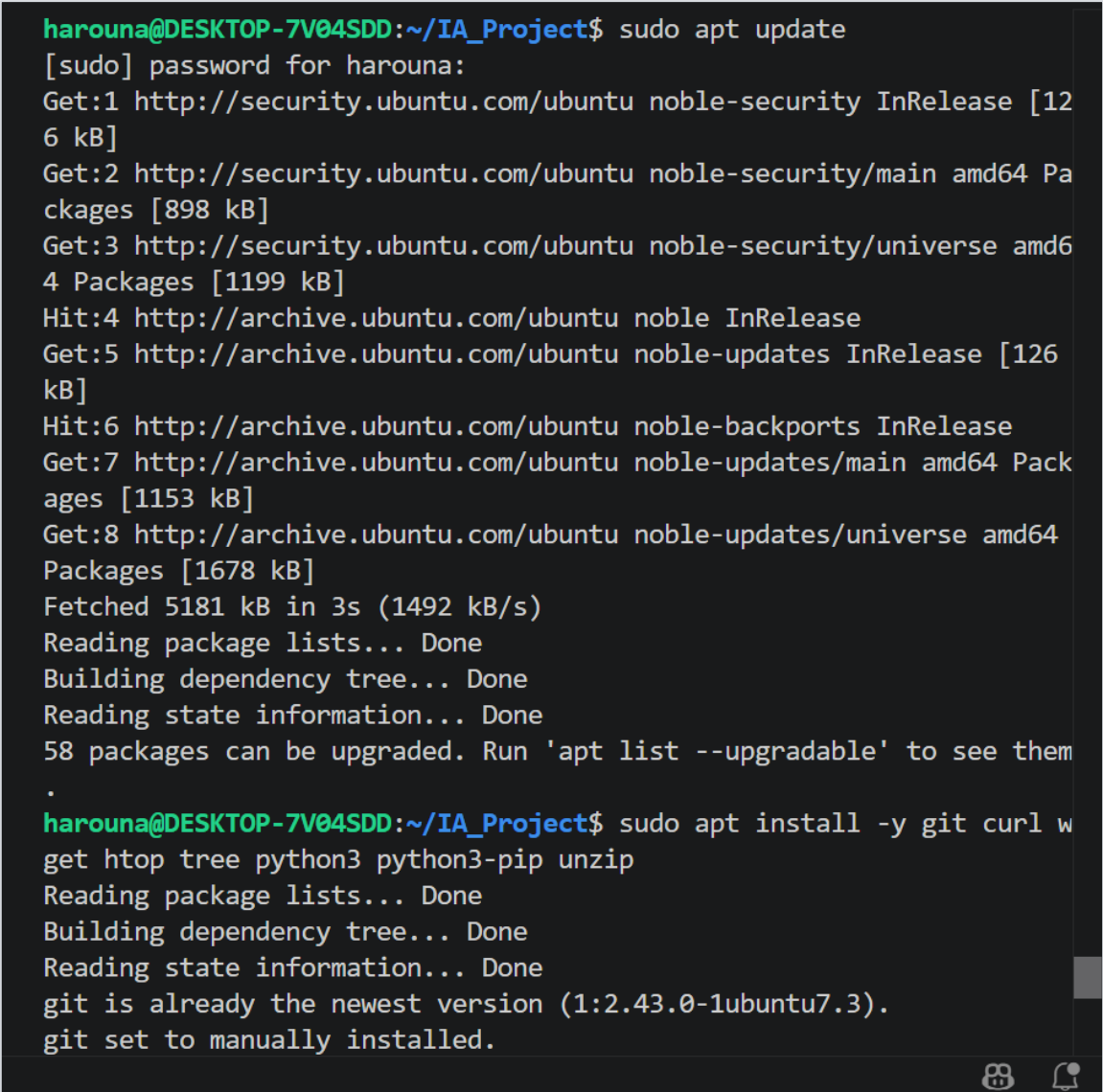
```
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo usermod -aG data,models frank
harouna@DESKTOP-7V04SDD:~/IA_Project$ id frank
uid=1012(frank) gid=1012(frank) groups=1012(frank),100(users),1007(data),1008(models)
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u frank bash -c 'touch ~harouna/IA_Project/backup/test.txt'
> ^C
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u frank bash -c 'touch ~harouna/IA_Project/datasets/frank_notes.txt && echo "O : frank écrit dans datasets"'
O : frank écrit dans datasets
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u frank bash -c 'touch ~harouna/IA_Project/datasets/frank_notes.txt && echo "OK : frank écrit dans datasets"'
OK : frank écrit dans datasets
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u frank touch ~harouna/IA_Project/backup/test.txt
touch: cannot touch '/home/harouna/IA_Project/backup/test.txt': Permission denied
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo -u frank bash -c 'cd ~harouna/IA_Project/shared && touch frank_shared.txt && ls -l'
total 0
-rw-rw-r-- 1 frank frank 0 Jul 28 17:17 frank_shared.txt
harouna@DESKTOP-7V04SDD:~/IA_Project$
```

Capture 21 — Création de frank et vérification de ses droits (a à e)

Partie 4 — Installation des logiciels

1) Installation

```
sudo apt update
sudo apt install -y git curl wget htop tree python3 python3-pip unzip
```

A terminal window screenshot showing the execution of two apt commands. The first command is 'sudo apt update', which updates the package lists from various sources including security.ubuntu.com and archive.ubuntu.com. It shows progress bars for downloading packages like noble-security InRelease, noble-security/main amd64 Packages, and noble-security/universe amd64 Packages. The second command is 'sudo apt install -y git curl wget htop tree python3 python3-pip unzip'. It shows that git is already the newest version and is set to manually installed, while the other packages are being installed. The terminal output is as follows:

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo apt update
[sudo] password for harouna:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [12
6 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security/main amd64 Pa
ckages [898 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/universe amd6
4 Packages [1199 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126
kB]
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Pack
ages [1153 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64
Packages [1678 kB]
Fetched 5181 kB in 3s (1492 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
58 packages can be upgraded. Run 'apt list --upgradable' to see them
.
harouna@DESKTOP-7V04SDD:~/IA_Project$ sudo apt install -y git curl w
get htop tree python3 python3-pip unzip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
```

Capture 22 — Installation des paquets (apt install)

2) Vérification

```
for o in git curl wget htop tree python3 pip3 unzip; do
    printf "%-10s : %s\n" "$o" "$(command -v $o || echo MANQUANT)"
done
git --version; python3 --version; pip3 --version; tree --version
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project$ for outil in git curl wget htop  
p tree python3 pip3 unzip; do  
    if command -v "$outil" > /dev/null 2>&1; then  
        printf "%-10s : OK -> %s\n" "$outil" "$(command -v $outil)"  
    else  
        printf "%-10s : MANQUANT\n" "$outil"  
    fi  
done  
git      : OK -> /usr/bin/git  
curl     : OK -> /usr/bin/curl  
wget     : OK -> /usr/bin/wget  
htop     : OK -> /usr/bin/htop  
tree     : OK -> /usr/bin/tree  
python3  : OK -> /usr/bin/python3  
pip3     : OK -> /usr/bin/pip3  
unzip    : OK -> /usr/bin/unzip  
harouna@DESKTOP-7V04SDD:~/IA_Project$
```

Capture 23 — Vérification des outils installés et de leurs versions

3) Téléchargement du dataset

```
cd ~/IA_Project/datasets/brut  
wget https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv
```

```
alice@DESKTOP-7V04SDD:/home/harouna/IA_Project$ cd datasets/brut
alice@DESKTOP-7V04SDD:/home/harouna/IA_Project/datasets/brut$ wget https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv
--2026-07-28 21:57:12-- https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3858 (3.8K) [text/plain]
Saving to: 'iris.csv'

iris.csv      100%[=====>]   3.77K  --.-KB/s    in 0s

2026-07-28 21:57:13 (18.0 MB/s) - 'iris.csv' saved [3858/3858]

alice@DESKTOP-7V04SDD:/home/harouna/IA_Project/datasets/brut$
```

Capture 24 — Téléchargement de iris.csv (wget)

4) Vérification de sa présence | 5) Consultation du contenu

```
ls -lh iris.csv
head -n 10 iris.csv
wc -l iris.csv
cut -d',' -f5 iris.csv | tail -n +2 | sort | uniq -c
```

```
alice@DESKTOP-7V04SDD:/home/harouna$ ls -lh ~/IA_Project/datasets/brut/iris.csv
ls: cannot access '/home/alice/IA_Project/datasets/brut/iris.csv': No such file or directory
alice@DESKTOP-7V04SDD:/home/harouna$ ls -lh IA_Project/datasets/brut/iris.csv
-rw-rw-r-- 1 alice alice 3.8K Jul 28 21:57 IA_Project/datasets/brut/iris.csv
alice@DESKTOP-7V04SDD:/home/harouna$ file IA_Project/datasets/brut/iris.csv
IA_Project/datasets/brut/iris.csv: CSV ASCII text
alice@DESKTOP-7V04SDD:/home/harouna$
```

```
alice@DESKTOP-7V04SDD:/home/harouna$ cd IA_Project/datasets/brut
alice@DESKTOP-7V04SDD:/home/harouna/IA_Project/datasets/brut$ head -n 10 iris.csv
sepal_length,sepal_width,petal_length,petal_width,species
5.1,3.5,1.4,0.2,setosa
4.9,3.0,1.4,0.2,setosa
4.7,3.2,1.3,0.2,setosa
4.6,3.1,1.5,0.2,setosa
5.0,3.6,1.4,0.2,setosa
5.4,3.9,1.7,0.4,setosa
4.6,3.4,1.4,0.3,setosa
5.0,3.4,1.5,0.2,setosa
4.4,2.9,1.4,0.2,setosa
alice@DESKTOP-7V04SDD:/home/harouna/IA_Project/datasets/brut$
```

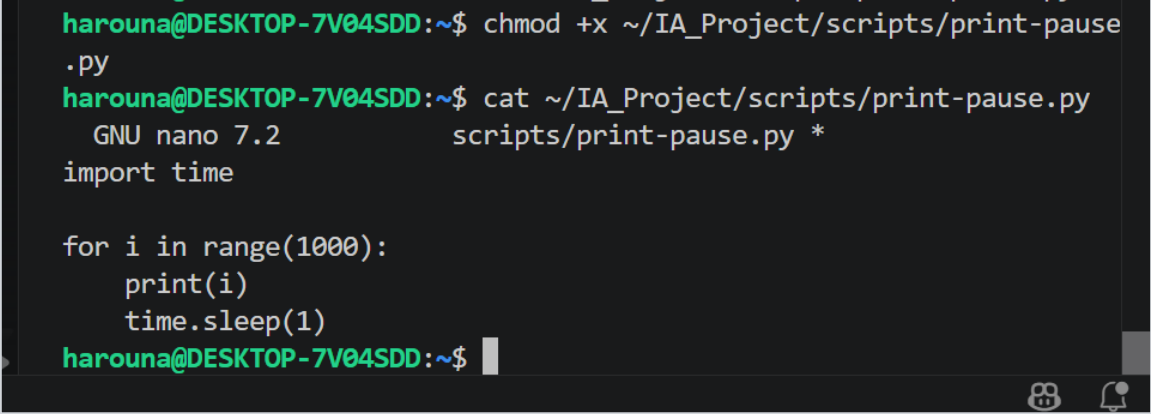
Capture 25 — Contenu et statistiques du dataset iris.csv

Partie 5 — Gestion des processus

1) Création du script print-pause.py

```
cat > ~/IA_Project/scripts/print-pause.py << 'EOF'
import time

for i in range(1000):
    print(i)
    time.sleep(1)
EOF
```



A terminal window showing the creation and editing of the script. The user runs `chmod +x ~/IA_Project/scripts/print-pause.py` and then `cat ~/IA_Project/scripts/print-pause.py`. The output shows the script content: `import time`, `for i in range(1000):`, `print(i)`, `time.sleep(1)`. The terminal prompt is `harouna@DESKTOP-7V04SDD:~$`.

Capture 26 — Script print-pause.py

2) Lancement | 3) Observation du PID

```
cd ~/IA_Project/scripts
python3 print-pause.py

# depuis un second terminal
pgrep -af print-pause
ps aux | grep print-pause
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project/scripts$ python3 print-pause.py
File "/home/harouna/IA_Project/scripts/print-pause.py", line 1
    GNU nano 7.2          scripts/print-pause.py *
```

```
IndentationError: unexpected indent
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project/scripts$ nano print-pause.py
```

```
harouna@DESKTOP-7V04SDD:~/IA_Project/scripts$ python3 print-pause.py
```

```
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
```

```
harouna@DESKTOP-7V04SDD:~$ ps aux | grep print-pause
harouna   21081  0.0  0.1 15340 10196 pts/2    S+   23:45
           0:00 python3 print-pause.py
harouna   21150  0.0  0.0  4096  1960 pts/4    S+   23:47
           0:00 grep --color=auto print-pause
harouna@DESKTOP-7V04SDD:~$ pgrep -af print-pause
21081 python3 print-pause.py
```

Capture 27 — Exécution du script et identification du PID

4) Utilisation CPU

```
PID=$(pgrep -f print-pause | head -n 1)
ps -p $PID -o pid,ppid,user,%cpu,%mem,etime,stat,cmd
top -p $PID
htop -p $PID
```

```

harouna@DESKTOP-7V04SDD:~$ PID=$(pgrep -f print-pause | head
-n 1)
harouna@DESKTOP-7V04SDD:~$ echo "PID = $PID"
PID = 21081
harouna@DESKTOP-7V04SDD:~$ ps -p "$PID" -o pid,ppid,user,%cpu
,%mem,etime,stat,cmd
    PID    PPID USER      %CPU %MEM     ELAPSED STAT CMD
    21081   20978 harouna    0.0  0.1         04:45 S+  python3 p
harouna@DESKTOP-7V04SDD:~$

```

Capture 28 — Consommation CPU du processus (ps / top / htop)

5) Arrêt du processus

```

kill $PID
kill -9 $PID      # si nécessaire
pkill -f print-pause
ps aux | grep print-pause

```

```

harouna@DESKTOP-7V04SDD:~$ ps -p "$PID" -o pid,ppid,user,%cpu
,%mem,etime,stat,cmd
    PID    PPID USER      %CPU %MEM     ELAPSED STAT CMD
    21081   20978 harouna    0.0  0.1         04:45 S+  python3 p
harouna@DESKTOP-7V04SDD:~$
harouna@DESKTOP-7V04SDD:~$ kill "$PID"
harouna@DESKTOP-7V04SDD:~$

```

```

352
353
354
Terminated
harouna@DESKTOP-7V04SDD:~/IA_Project/scripts$

```

Capture 29 — Arrêt du processus (kill)

6) Relance en arrière-plan | 7) Retour au premier plan

```

python3 print-pause.py &
jobs -l
echo "PID : $!"

# Ctrl+Z pour suspendre
bg %1
fg %1

```

```
harouna@DESKTOP-7V04SDD:~/IA_Project/scripts$ python3 print-p  
ause.py &  
[1] 22671
```

```
harouna@DESKTOP-7V04SDD:~$ echo "PID du dernier job : $!"  
PID du dernier job : 22665  
harouna@DESKTOP-7V04SDD:~$ python3 print-pause.py > ~/IA_Proj  
ect/logs/pause.log 2>&1 &  
tail -f ~/IA_Project/logs/pause.log  
[1] 22672
```

```
[1]+  Running                  python3 print-pause.py &  
harouna@DESKTOP-7V04SDD:~/IA_Project/scripts$ 24
```

Capture 30 — Gestion arrière-plan / premier plan (&, jobs, bg, fg)

Partie 6 — Bash scripting : setup_project.sh

Le script automatise : saisie du nom du projet, création de l'arborescence, génération du fichier de configuration, installation des outils, téléchargement du dataset, compression et affichage du résumé.

Structure du script

```
#!/bin/bash
# =====
# setup_project.sh - Creation automatique d'un projet IA
# Formation IA - Promo 1 - Sonatel Academie
# =====

# 1. Demander le nom du projet
read -p "Nom du projet : " NOM
PROJET="$HOME/$NOM"

# 2. Creer l'arborescence
mkdir -p
$PROJET/{datasets/brut,datasets/clean,config,logs,scripts,models,api,backup,documentation,shared}

# 3. Creer le fichier de configuration
cat > $PROJET/config/settings.conf << EOF
PROJECT_NAME=$NOM
DATA_PATH=datasets/brut
MODEL_PATH=models
LOG_LEVEL=INFO
API_PORT=8000
AUTHOR=$(whoami)
EOF

# 4. Installer les outils
sudo apt install -y git curl wget htop tree python3 python3-pip unzip

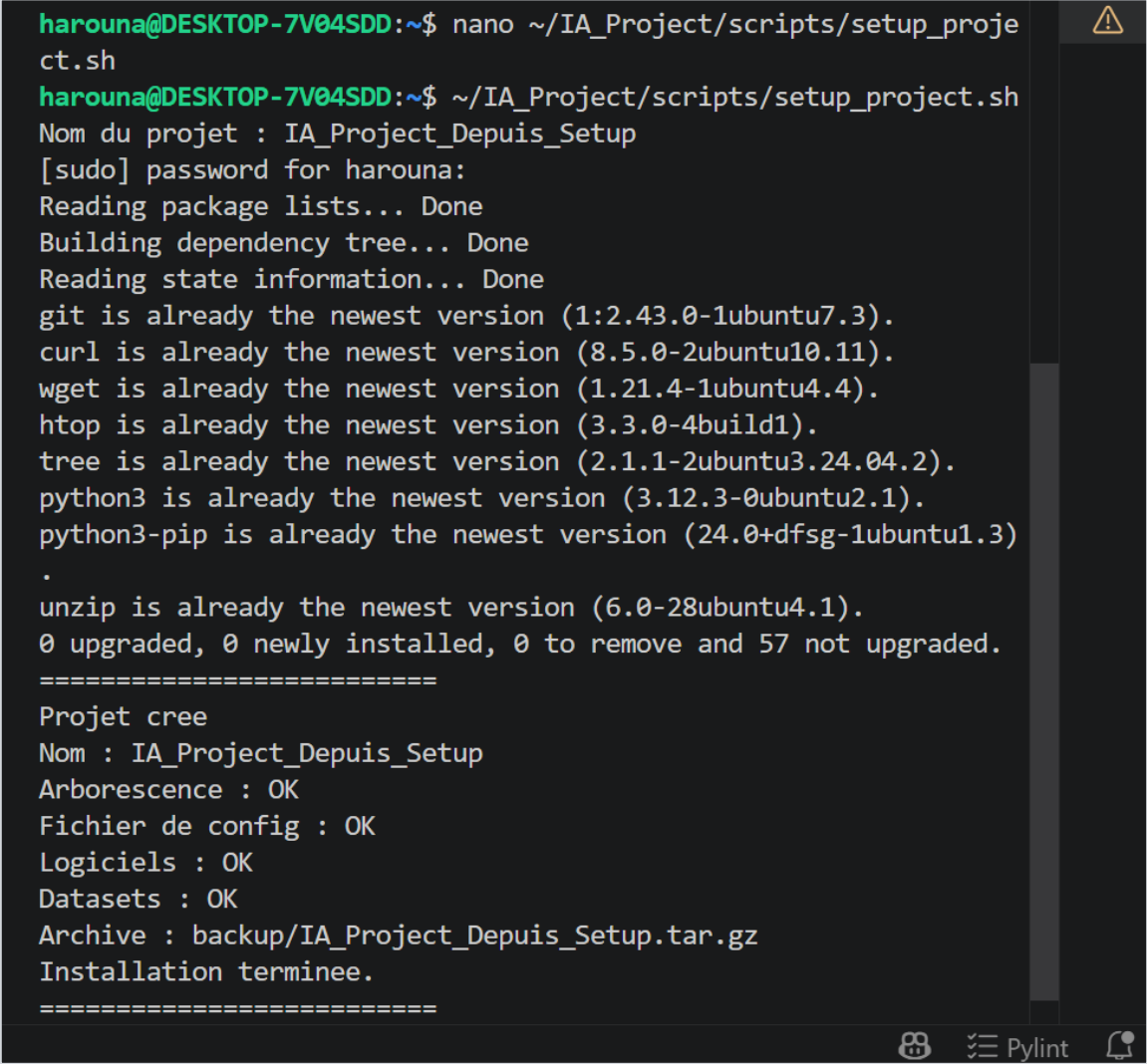
# 5. Telecharger le dataset
wget -q -O $PROJET/datasets/brut/iris.csv \
  https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv

# 6. Compresser le projet
tar -czf $PROJET/backup/$NOM.tar.gz -C $HOME --exclude="$NOM/backup" $NOM

# 7. Afficher le resume
echo "======"
echo "Projet cree"
echo "Nom : $NOM"
echo "Arborescence : OK"
echo "Fichier de config : OK"
echo "Logiciels : OK"
echo "Datasets : OK"
echo "Archive : backup/$NOM.tar.gz"
echo "Installation terminee."
echo "======"
```

Exécution

```
chmod +x ~/IA_Project/scripts/setup_project.sh
bash -n ~/IA_Project/scripts/setup_project.sh
~/IA_Project/scripts/setup_project.sh
```



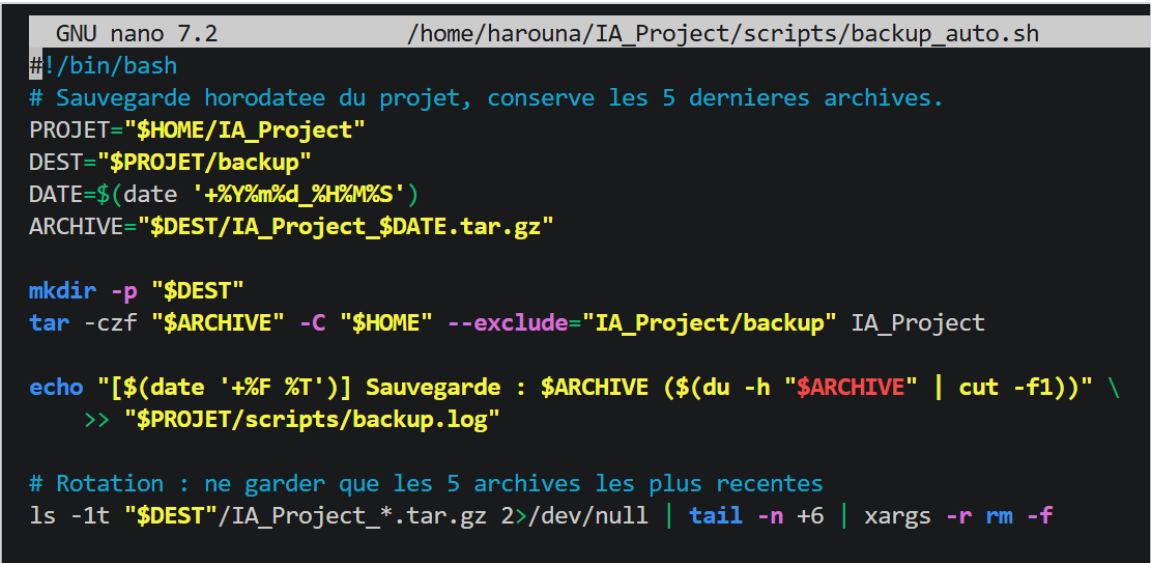
```
harouna@DESKTOP-7V04SDD:~$ nano ~/IA_Project/scripts/setup_proje
ct.sh
harouna@DESKTOP-7V04SDD:~$ ~/IA_Project/scripts/setup_project.sh
Nom du projet : IA_Project_Depuis_Setup
[sudo] password for harouna:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
curl is already the newest version (8.5.0-2ubuntu10.11).
wget is already the newest version (1.21.4-1ubuntu4.4).
htop is already the newest version (3.3.0-4build1).
tree is already the newest version (2.1.1-2ubuntu3.24.04.2).
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3-pip is already the newest version (24.0+dfsg-1ubuntu1.3).
.
unzip is already the newest version (6.0-28ubuntu4.1).
0 upgraded, 0 newly installed, 0 to remove and 57 not upgraded.
=====
Projet cree
Nom : IA_Project_Depuis_Setup
Arborescence : OK
Fichier de config : OK
Logiciels : OK
Datasets : OK
Archive : backup/IA_Project_Depuis_Setup.tar.gz
Installation terminee.
=====
```

Capture 32 — Exécution du script et résumé final affiché

Partie 7 — Bonus

Bonus Sauvegarde automatique horodatée (cron)

```
~/IA_Project/scripts/backup_auto.sh
crontab -e      # 0 20 * * * /home/$USER/IA_Project/scripts/backup_auto.sh
crontab -l
```



```
GNU nano 7.2 /home/harouna/IA_Project/scripts/backup_auto.sh
#!/bin/bash
# Sauvegarde horodatee du projet, conserve les 5 dernieres archives.
PROJET="$HOME/IA_Project"
DEST="$PROJET/backup"
DATE=$(date '+%Y%m%d_%H%M%S')
ARCHIVE="$DEST/IA_Project_$DATE.tar.gz"

mkdir -p "$DEST"
tar -czf "$ARCHIVE" -C "$HOME" --exclude="IA_Project/backup" IA_Project

echo "[$(date '+%F %T')] Sauvegarde : $ARCHIVE ($(du -h "$ARCHIVE" | cut -f1))" \
>> "$PROJET/scripts/backup.log"

# Rotation : ne garder que les 5 archives les plus recentes
ls -lt "$DEST"/IA_Project_*.tar.gz 2>/dev/null | tail -n +6 | xargs -r rm -f
```

Capture 35 — Sauvegarde horodatée et planification cron

Livrables

N°	Livrable	Emplacement
1	Arborescence complète du projet	~/IA_Project
2	Utilisateurs, groupes et permissions	/etc/passwd, /etc/group, ls -l
3	Logiciels installés et vérifiés	command -v, --version
4	Dataset téléchargé	datasets/brut/iris.csv
5	Archive de sauvegarde	backup/projet_IA.tar.gz
6	Script setup_project.sh commenté	scripts/setup_project.sh
7	Document des commandes Linux	documentation/COMMANDES.md

Tableau 2 — Récapitulatif des livrables de l'atelier

Conclusion

L'atelier a permis de mettre en place un environnement Linux complet pour un projet IA : structuration des données, sécurisation par utilisateurs, groupes et permissions, installation de l'outillage, récupération d'un dataset, supervision des processus, sauvegarde et automatisation par script Bash.

Ces compétences constituent le socle du déploiement de modèles en production, où les traitements s'exécutent sur des serveurs Linux sans interface graphique.

Principales commandes utilisées

Domaine	Commandes
Arborescence	mkdir -p, touch, tree, cd, pwd
Lecture	cat, head, tail, less, wc -l, cut, sort, uniq
Recherche	grep -rn, find -name, find -type f
Fichiers	cp, mv, rm, du, ls -lhS, >, >>,
Archivage	tar -czvf, tar -xzvf, unzip
Utilisateurs	adduser, useradd, passwd, id, groups, su, sudo -u
Groupes	groupadd, usermod -aG
Permissions	chmod, chown, chgrp, stat, umask, namei
Paquets	apt update, apt install, dpkg -l, command -v, pip
Réseau	wget, curl
Processus	ps aux, pgrep, top, htop, kill, pkill, jobs, bg, fg, &

Tableau 3 — Principales commandes Linux utilisées